

Summer Workshop: Causal Inference for Uber Pool

From observed operational data to a causal decision model

Business Case and Causal Question

Uber tested a policy change in Uber Pool: allowing riders to wait up to **5 minutes** instead of **2 minutes** before matching. The operational intuition is that a longer waiting window may improve shared matching, increase the number of pooled rides, and reduce matching costs. However, the policy may also increase rider cancellations or change the type of trips completed.

The central question is not predictive. The goal is not only to ask whether treated periods have higher or lower payouts. The causal question is:

What is the causal effect of increasing the Uber Pool waiting window from 2 minutes to 5 minutes on operational performance?

The main estimand is the average treatment effect:

$$ATE = \mathbb{E}[Y(1) - Y(0)],$$

where $Y(1)$ is the outcome under the 5-minute waiting policy and $Y(0)$ is the outcome under the 2-minute waiting policy.

Learning Objectives

By the end of the workshop, each team should be able to:

- distinguish association, correlation, and causality in the Uber Pool setting;
- identify the treatment, outcome, contextual variables, mediators, and possible post-treatment variables;
- construct a causal DAG that matches a specific business question;
- explain why adding more controls can improve prediction but damage causal interpretation;
- propose a defensible causal model and communicate it as a decision recommendation.

Data Dictionary

Workshop Tasks

Part 1. Define the causal problem

Answer the following questions before drawing any model:

1. What is the intervention? Write it as a **do**-operation: $do(X = x)$.
2. What is the main outcome? Is `total_driver_payout` a benefit, a cost, or an operational performance measure?
3. What is the main business decision: keep the 5-minute policy, return to 2 minutes, or use it only in specific contexts?
4. Which variables are observed before or at the time of treatment, and which variables may happen after treatment?

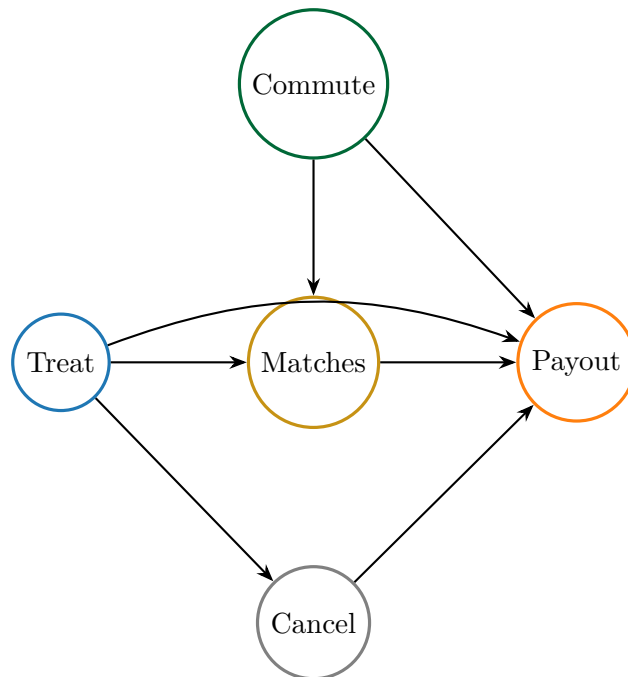
Variable	Type	Definition
<code>period_start</code>	Time	Time period in which the data were gathered.
<code>wait_time</code>	String	Matching policy: <code>2 mins</code> if riders could wait up to 2 minutes; <code>5 mins</code> if riders could wait up to 5 minutes.
<code>treat</code>	Boolean	Treatment indicator: <code>TRUE</code> when <code>wait_time = 5 mins</code> ; <code>FALSE</code> when <code>wait_time = 2 mins</code> .
<code>commute</code>	Boolean	Rush-hour indicator: <code>TRUE</code> during commuting periods, approximately 7:00–9:40 AM or 3:00–5:40 PM; <code>FALSE</code> otherwise.
<code>trips_pool</code>	Numeric	Total number of completed Pool trips in the current period. Each matched ride request is counted as a separate trip.
<code>trips_express</code>	Numeric	Total number of completed Express Pool trips in the current period. Each matched ride request is counted as a separate trip.
<code>rider_cancellations</code>	Numeric	Total number of requested trips cancelled by riders in the current period.
<code>total_driver_payout</code>	Numeric	Total pesos paid to drivers for completed trips in the current period; this represents Uber’s total matching cost.
<code>total_matches</code>	Numeric	Number of completed trips paired with at least one other rider for part of the trip. Each matched ride request is counted separately.
<code>total_double_matches</code>	Numeric	Number of completed trips paired with at least two other riders for part of the trip. Each matched ride request is counted separately.

Part 2. Build a first causal DAG

Construct a DAG that includes at least the following nodes:

`treat`, `commute`, `total_matches`, `rider_cancellations`, `total_driver_payout`.

A suggested starting structure is:



Discuss whether `commute` is a context variable, a confounder, an effect modifier, or all of these. Then decide whether `total_matches` and `rider_cancellations` are mediators, outcomes, colliders, or variables that should be handled carefully.

Part 3. Compare three causal models

Your team must compare the meaning of three candidate models. The goal is not to choose the “best fitting” predictive model, but to explain exactly what causal question each model answers.

1. Naive association model

$$\text{total_driver_payout}_i = \beta_0 + \beta_1 \text{treat}_i + \varepsilon_i.$$

Discussion Prompt: What can this model say? What can it not say? Consider that treatment (the 5-minute window) might have been tested mostly during rush hours (commute times) when payouts are naturally higher due to traffic. If that is true, what happens to β_1 ?

2. Adjusted total-effect model

$$\text{total_driver_payout}_i = \beta_0 + \beta_1 \text{treat}_i + \beta_2 \text{commute}_i + \varepsilon_i.$$

Discussion Prompt: Under what assumptions can β_1 be interpreted as the *total policy effect*? Why is it important to control for `commute` (rush hour)? Does controlling for commute block any part of the actual treatment effect, or does it merely block a spurious backdoor path?

3. Mechanism or direct-effect model

$$\text{total_driver_payout}_i = \beta_0 + \beta_1 \text{treat}_i + \beta_2 \text{commute}_i + \beta_3 \text{total_matches}_i + \varepsilon_i.$$

Discussion Prompt: Why does adding `total_matches` change the question from “What is the total effect of the policy?” to “What is the direct effect of the policy, assuming the matching rate stayed exactly the same?” If the entire goal of the 5-minute wait is to increase matches to lower costs, does this model help Uber understand if the policy worked as intended?

Part 4. Identify risks of causal misinterpretation

Prepare one clear explanation of each risk. To help you, here are deeper context clues for the Uber Pool experiment:

- **Confounding (Omitted Variable Bias):** A third variable affects both treatment assignment and the outcome. *Example in Uber:* If the 5-minute waiting policy was only rolled out during heavy rain or peak rush hours, the weather or time of day is a confounder. If you don’t control for it, you might falsely conclude the 5-minute wait **causes** longer trips and higher payouts.
- **Mediation (Over-control Bias):** Part of the treatment effect flows through an intermediate variable. *Example in Uber:* The policy causes riders to wait longer, which allows the algorithm to find more `total_matches`, which subsequently reduces the `total_driver_payout`. If you control for matches in your regression, you statistically delete the exact mechanism that makes the policy successful!
- **Collider Bias (Berkson’s Paradox):** Conditioning on a variable jointly influenced by two independent variables. *Example in Uber:* Imagine you only analyze trips where `rider_cancellations` > 0. Both the “5-minute wait” (treatment) and “Traffic Jams” (unobserved variable) cause cancellations. By restricting your analysis to cancelled trips, you create a fake mathematical correlation between the treatment and traffic jams.
- **Post-treatment Bias:** Controlling for any variable that was measured *after* the treatment was applied. *Example in Uber:* Controlling for the final `trips_pool` completed. Because the policy directly affects how many trips are completed, including it in the model absorbs the effect of the policy, hiding the true causal impact.

Final Deliverable: 7-Minute Presentation

Each group must prepare a short presentation of **maximum 7 minutes**. There is **no written report**. The presentation should include no more than five slides:

1. **Business question:** What decision should Uber make about the 5-minute Pool waiting policy?
2. **Data and variables:** Identify treatment, outcome, context variables, mediators, and possible post-treatment variables.
3. **DAG:** Present your causal graph and explain the main pathways.
4. **Model choice:** Compare the naive, adjusted, and mechanism models. State which one answers your main question.
5. **Decision recommendation:** Should Uber scale, reject, or conditionally apply the policy? What evidence would you still need?